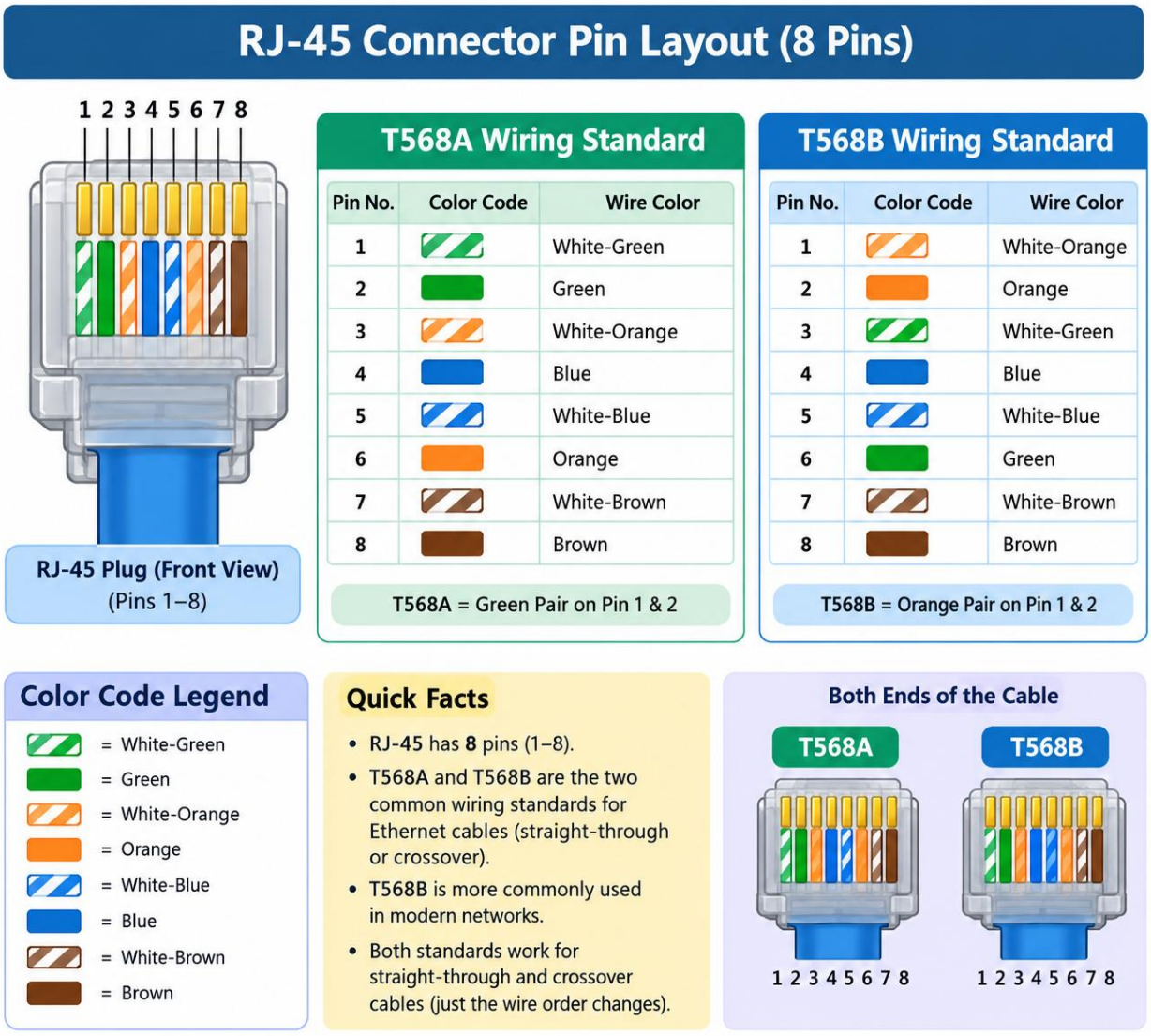
1.Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

Ans.



2.Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ans.

Specification	10BASE-T	100BASE-TX	1000BASE-T
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	Cat 3 or better twisted-pair cable	Cat 5 or better twisted-pair cable	Cat 5e or better twisted-pair cable
Maximum cable length	100 metres	100 metres	100 metres
Connector type	RJ-45 (8-pin)	RJ-45 (8-pin)	RJ-45 (8-pin)
Common name	Standard Ethernet	Fast Ethernet	Gigabit Ethernet
Wire pairs used	2 pairs	2 pairs	4 pairs

3.Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why. Think about how transmit and receive pairs are wired in each cable type.

Ans. To connect two laptops directly using an Ethernet cable, we traditionally use a crossover cable.

Why?

- **Crossover cable: It swaps the transmit (send) and receive pairs. This allows one laptop's transmit wires to connect to the other laptop's receive wires.**
- **Straight-through cable: It has the same wire color order at both ends and is normally used to connect different types of devices, like a laptop to a router.**

4.List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Ans.

Device	Likely Ethernet Standard	Ethernet Standard
1. Wi-Fi Router	1000BASE-T	Many modern routers support speeds up to 1 Gbps through an Ethernet cable.
2. Desktop Computer	1000BASE-T	Many desktop computers use Gigabit Ethernet for fast internet and file sharing.
3. Smart TV	100BASE-TX	Some smart TVs use Fast Ethernet to connect to the internet through a cable.